

Educational Background

University of Electronic Science and Technology of China

Sept. 2022 - Jun. 2026 (expected)

B.Eng in Communication Engineering

- Rank – 2/168 (top 1.2%) GPA – 3.92/4
- Honors: 2024 National Scholarship (Top 1.5%) for Undergraduates
- Language: IELTS: 6.5 CET-4: 602 CET-6: 602

Research Experiences

1. Metis: Memory Foundation Models

Jun. 2026 - Present

Project Description: Focused on the long-term memory capabilities of large language models, this project investigates the architectural design and training mechanisms of the memory foundation models, reproduces the training and evaluation pipeline based on Qwen-series models, and explores scalable memory architectures for long-term interaction tasks.

- **In-depth Analysis of Native Memory Mechanisms in LLMs:** Systematically studied the memory-native modeling paradigm and the core concepts of Native Memory States and Native Memory Procedures; designed the Local Memory and Hyper Memory modules, and implemented memory attention computation.
- **Design of an MoE-like Scalable Memory Architecture:** To address the fixed-capacity limitation of Local Memory in Metis, proposed a Mixture-of-Storage (MoS) extension that introduces a dynamic memory routing mechanism, enabling scalable memory capacity expansion while exploring approaches to improving long-term memory capabilities.
- **Construction of a General Capability Evaluation Framework for LLMs:** Designed general capability experiments based on Metis-V2.4-GDN and Qwen3.5-4B; conducted evaluations on benchmarks including MMLU-Pro, IFEval, and GSM8K; and analyzed the effects of native memory mechanisms on model reasoning and multilingual capabilities.

2. Lightweight Large Language Model Based on LLaMA Architecture

Nov. 2025 - Jan. 2026

Project Description: Aimed at lightweight LLM training scenarios, this project reproduces the complete experimental pipeline from model construction, pre-training, SFT, LoRA, to RLHF, practicing the full-chain pipeline for a 64M-parameter LLM.

- **Building a Lightweight LLM from Scratch:** Implemented a lightweight model based on the Decoder-only architecture, constructing core modules including RoPE positional encoding, KV Cache, GQA, RMSNorm, FFN, and SwiGLU, providing a unified model backbone for subsequent training and alignment.
- **Complete Pre-training and Fine-tuning Workflow:** Organized the dataset construction differences across training stages such as pre-training and SFT; reproduced the pre-training and post-training code pipelines, gaining hands-on experience with mixed-precision training, gradient accumulation, checkpoint saving, resume training, and distributed training; understood dialogue template design, label construction, and loss masking to improve training stability.
- **Post-training and Preference Alignment:** Implemented reinforcement-learning-based post-training methods including PPO, DPO, and GRPO using preference data and reward signals, mastering the core ideas of how models leverage preference data, rule-based rewards, and penalty constraints to enhance reasoning capability and reasoning stability.
- **Model Evaluation and Performance Analysis:** Reproduced the lightweight LLM evaluation pipeline on C-Eval and CMMLU benchmarks, compared accuracy changes across different checkpoints, and quantitatively analyzed the impact of each training stage on Chinese knowledge QA, commonsense reasoning, and instruction-following capabilities.

3. AI Programming Assistant System Based on the ReAct Paradigm

Feb. 2026 - Mar. 2026

Project Description: Built an AI Agent framework from scratch following the ReAct paradigm, addressing engineering challenges such as limited context windows, complex tool-calling chains, and multi-turn state maintenance, with features including multi-tool orchestration, context compression, session persistence, and long-term memory management.

- **Core Architecture Design:** Implemented an Agent loop of LLM reasoning → tool calling → result feedback, supporting multi-turn tool collaboration, output truncation recovery, and asynchronous task processing; designed a checkpoint snapshot mechanism to save states at critical tool-calling nodes, improving interrupt recovery and context consistency.
- **Plugin-style Tool System:** Built a tool registration architecture based on the Tool abstract class; new tools can be integrated simply by inheriting Tool and declaring a JSON Schema; supported concurrency-safe tiering and batch scheduling, with built-in dangerous command interception and directory sandbox restrictions in the Shell tool to reduce execution risks.
- **Tool Integration and Message Decoupling:** Implemented dynamic discovery and unified registration of external tools via the MCP Client-Server architecture, and designed an asynchronous MessageBus to decouple WebSocket interactions, the Agent main loop, and result distribution, facilitating future extensions for CLI, Cron, and third-party channels.

- **Hierarchical Memory System:** Maintained in-session dialogue context as short-term working memory with real-time token monitoring; triggered mid-term summary memory based on token thresholds—when token consumption exceeded limits, compressed historical dialogue context into structured summaries and persisted them; combined with a Dream mechanism to periodically organize cross-session long-term memory, accumulating user preferences, historical tasks, and stable facts, thereby enhancing the Agent’s cross-session continuity.

4. CNN-Based Multispectral and Panchromatic Image Fusion and Image Denoising Sept. 2024 - Dec. 2024

Project Description: Focused on remote sensing image fusion and image restoration, this project investigates convolutional-neural-network-based methods for fusing multispectral and panchromatic images and develops a ResNet-based image denoising network to effectively remove rain streaks, noise, and other visual artifacts.

- **Study of Multispectral and Panchromatic Image Fusion Algorithms:** Studied and analyzed representative convolutional neural network architectures, including PanNet, DiCNN, and FusionNet, and investigated feature extraction and fusion methods for integrating the spatial details of panchromatic images with the spectral information of multispectral images.
- **Design of a Residual-Network-Based Image Denoising Model:** Developed a CNN-based image restoration network using ResNet residual structures; leveraged residual feature learning to extract and suppress rain streaks and noise, enabling effective image denoising and fine-detail restoration.

5. Movable-Antenna-Enabled Near-Field Wireless Sensing Sept. 2024 - Jun. 2025

Project Description: Independently proposed and developed a movable-antenna-array-assisted near-field target localization framework, in which antenna positions are dynamically adjusted according to real-time channel state information extracted from echo signals. Compared with conventional uniform arrays, the proposed approach significantly reduces the Cramér–Rao bounds (CRBs) for angle and range estimation errors.

- **System Modeling and Theoretical Derivation:** Conducted the complete research workflow from literature review and near-field channel modeling to parameter estimation formulation, Fisher information matrix derivation, and theoretical performance analysis.
- **Simplified Theoretical Analysis and Algorithm Design:** Reformulated the complex joint estimation problem as a worst-case CRB minimization problem; derived globally optimal closed-form solutions for tractable cases and developed a low-complexity alternating optimization algorithm with carefully designed initialization strategies for more challenging cases, substantially improving algorithm performance.
- **Simulation Validation and Performance Analysis:** Conducted extensive simulations to evaluate the proposed movable antenna array and demonstrated a reduction of more than 50% in CRB compared with uniform antenna arrays.

Publications

- [C1] Y. Wang, W. Mei, X. Wei, B. Ning, and Z. Chen, “Antenna Position Optimization for Movable Antenna-Empowered Near-Field Sensing,” accepted by *ICC Workshops*, 2025, arXiv:2502.03169. [\[PDF\]](#)
- [J1] Y. Wang, W. Mei, X. Wei, B. Ning, and Z. Chen, “Movable Antenna Empowered Near-Field Sensing via Antenna Position Optimization,” submitted to *IEEE Transactions on Wireless Communications*. [\[PDF\]](#)